

Omar Elasfar

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EDUCATION

Bachelor of Science in Computer Science, Minor in Statistics

Expected Aug 2026

Carleton University

Ottawa, ON

- **CGPA:** 11.61/12.0 (A), *Fourth Year Standing*
- **Courses:** OOP, Operating Systems, Systems Programming, Software Engineering, Databases, Web Apps

SKILLS

Languages: Java, C/C++, Python, SQL, JavaScript, HTML, CSS, Bash, Go, R

Developer Tools: Git, Maven, Docker, Kubernetes, Cloud Services (AWS), TeamCity, Linux/Unix, Jira

Competencies: Verbal and Written Communication Skills, Detail Oriented, Problem-Solver, Initiative, Leadership

EXPERIENCE

Software Developer

May 2025 – Dec 2025, Apr 2026 – Present

Ciena

Kanata, ON

- Automated **concurrent** concatenation of client parquet files stored in **AWS S3** using **Java** & **Maven** for development, **Kubernetes** for testing, and **Docker** for deployment, ensuring fast and reliable data aggregation.
- Packaged a **Maven**-based **Java** service into a Debian package and built **Linux** startup scripts and systemd unit files to enable secure, reliable background execution and lifecycle management.
- Enhanced the SaaS platform **UI** by implementing pagination, a smart search box, and advanced **SQL** queries using **Java**, **PostgreSQL**, and **JavaScript**, improving **user experience** and system loading times by **75%**.
- Validated and deployed system upgrades by developing **JUnit** unit tests, running containerized builds in **Docker**, and executing **CI/CD pipelines** in **TeamCity**, strengthening **software reliability** and deployment consistency.

Full Stack Software Developer

Jan 2025 – Apr 2025

Versaterm Public Safety Inc.

Ottawa, ON

- Developed a centralized Event Locations feature to streamline event-related data management across the U.S. Record Management System (RMS), utilizing Genero 4GL, **SQL**, and Handlebars.
- Built and deployed **40+** database views across **PostgreSQL**, Informix, and **SQL Server** engines to support a major schema overhaul, normalize database structures, and maintain client compatibility.
- Diagnosed, **troubleshooted**, and resolved system issues while collaborating with cross-functional teams via **Jira**, **Confluence**, and **Git**, documenting technical solutions and contributing to **agile** workflows.

Mission Scientist

Jan 2025 – Apr 2025

Canadian Space Agency

Remote

- Developed an anomaly detection **algorithm** for satellite river data to identify flood and drought events triggered by landslides, using statistical confidence intervals and time-series smoothing techniques in **R**.
- Optimized and enhanced legacy **Python code** for cleaning satellite data by **automating** data validation and transformation, achieving an **80% increase** in data processing efficiency.

Test Automation Engineer

Sep 2024 – Apr 2025

Qurani

Remote

- Conducted rigorous **black-box testing** of an educational **Android app** by performing tests across multiple devices and versions, reporting **15+ bugs** to developers, resulting in improved functionality and reliability.
- Designed and implemented **test cases** by developing **manual** and **automated testing** procedures using **Python frameworks**, increasing test coverage and enhancing the application's overall reliability.

Computer Science Teaching Assistant

Sep 2024 – Dec 2024

Carleton University

Ottawa, ON

- Guided **50+** students in mastering **software engineering** by offering personalized office hours, detailed feedback, and interactive labs, resulting in a **50%** increase in coding proficiency.
- Assisted students in **troubleshooting** and optimizing **Python** codebases through targeted guidance and problem-solving strategies, prompting a **70%** reduction in error rates.

PROJECTS

Real-Time ML-Based Network Anomaly Detector | C, QNX, Machine Learning

Mar 2026 – Apr 2026

- Developed a **real-time network anomaly detection system** in **C** on the **QNX RTOS** to monitor live 5G/6G-style traffic and detect abnormal behavior using a neural-network **autoencoder**.
- Implemented a controller-driven pipeline for **metric collection**, **data despiking**, **model training**, and **inference** with deterministic scheduling using **POSIX threads** and **round-robin scheduling**.
- Achieved **sub-300 ms anomaly detection latency**, exceeding the project target of under 1 second while maintaining predictable periodic execution and deadline monitoring.

Hack-the-Tunnels 2024 | TypeScript, React, Prisma, Express

Oct 2024